LAB 10 - Superposition Theorem

TECH 60 - Introduction to Electronics - Sec 14
F26
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Objective

The objective of this lab is to understand the Superposition Theorem through hand calculation, Multisim simulation, and a hardwired breadboard circuit. We find the voltage across R2 caused by each source acting alone (V2,S1 and V2,S2) and show that their sum equals the voltage across R2 in the original circuit with both sources on.

Equipment/Materials

- Solderless breadboard
- Keysight EDU36311A Triple Output Programmable DC Power Supply
- Keysight EDU34450A 5½ Digit Digital Multimeter (DMM)
- Multisim software running on PC
- 390 Ω resistor (Quantity 1)
- 470 Ω resistor (Quantity 1)
- 750 Ω resistor (Quantity 1)
- Jumper wires

Part I. Hand Calculations

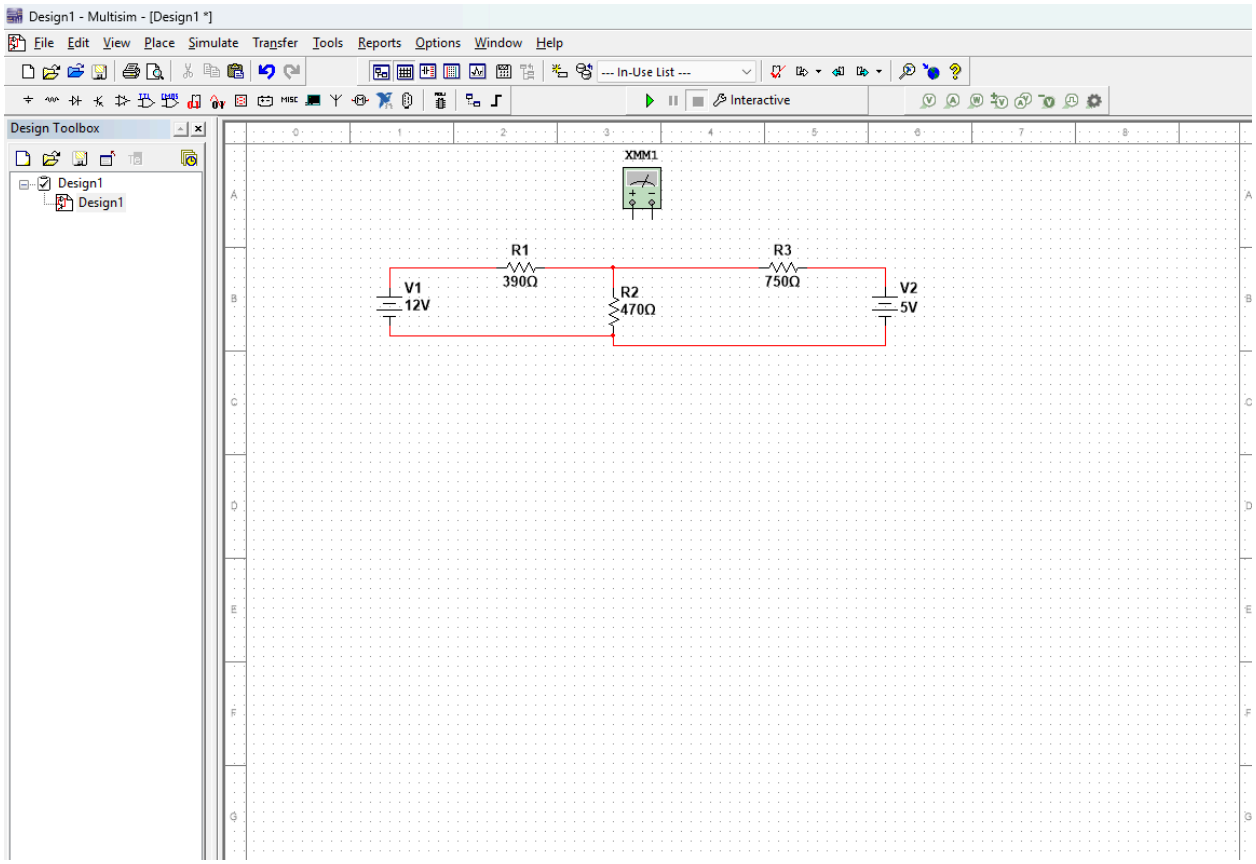
Given: Vs1 = 12 V, Vs2 = 5 V, R1 = 390 Ω, R2 = 470 Ω, R3 = 750 Ω
Step 1 - Vs1 only (Vs2 replaced with a short circuit). R2 and R3 are in parallel, and (R2 || R3) is in series with R1
R2||R3 = (R2 × R3)/(R2 + R3) = (470 × 750)/(470 + 750) = 352500/1220 = 288.93 Ω
V2,S1 = Vs1 × (R2||R3)/(R1 + R2||R3) = 12 V × 288.93 Ω/(390 Ω + 288.93 Ω) = **5.107 V**

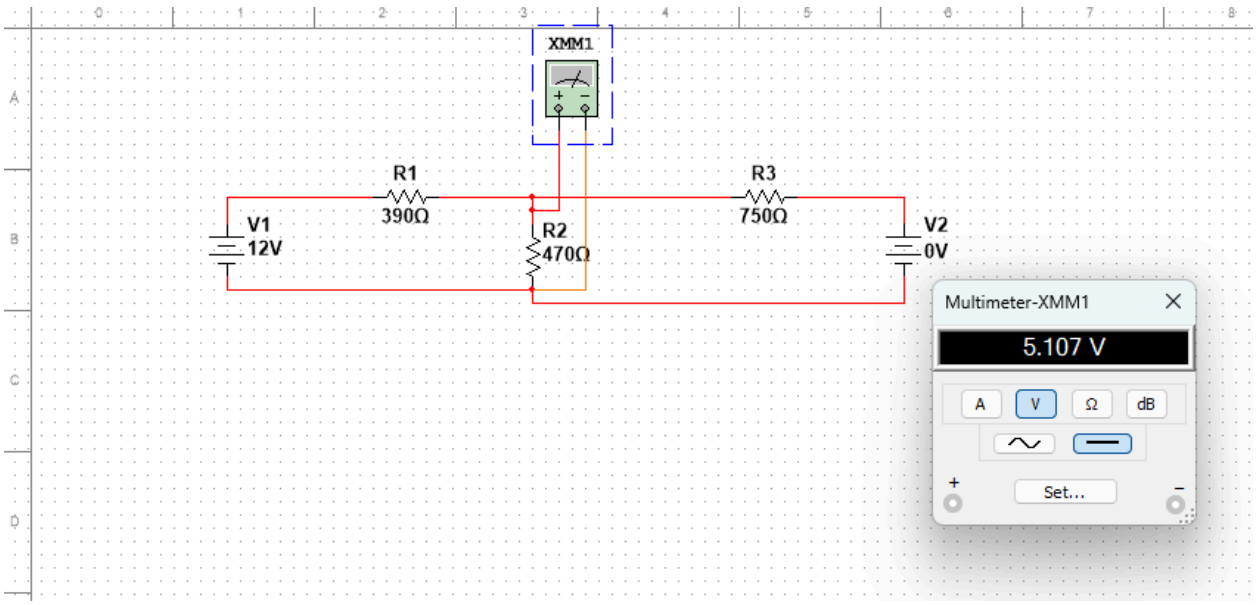
Step 2 - Vs2 only (Vs1 replaced with a short circuit). R1 and R2 are in parallel, and that combination is in series with R3.
R1||R2 = (R1 × R2)/(R1 + R2) = (390 × 470)/(390 + 470) = 183300/860 = 213.14 Ω
V2,S2 = Vs2 × (R1||R2)/(R3 + R1||R2) = 5 V × 213.14 Ω/(750 Ω + 213.14 Ω) = **1.106 V**

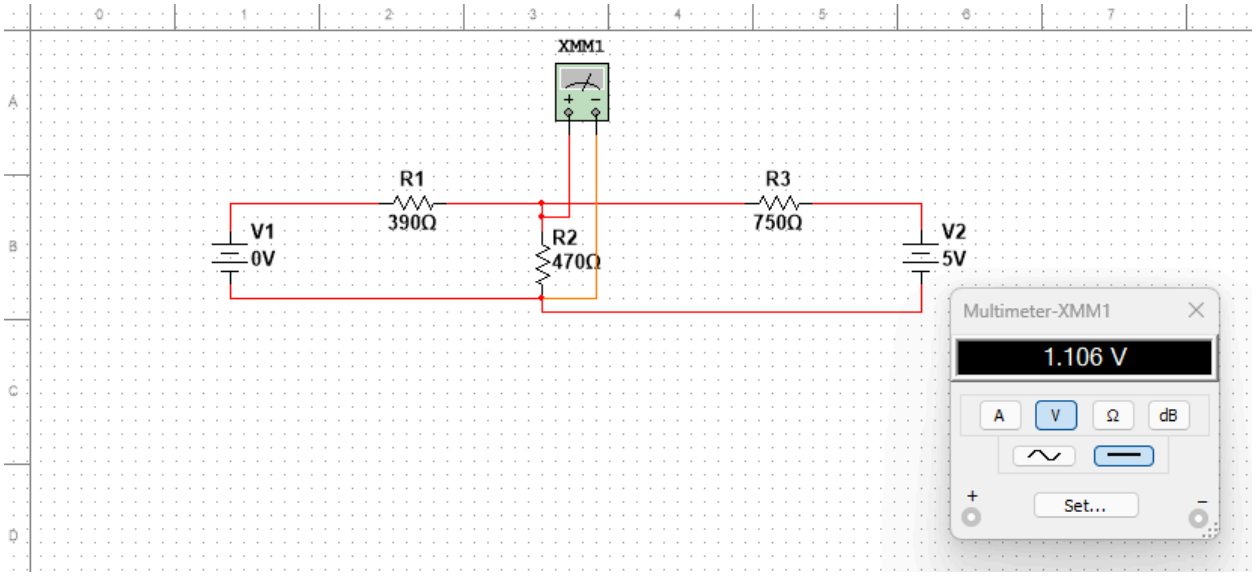
Step 3 - Superposition.
V2 = V2,S1 + V2,S2 = 5.107 V + 1.106 V = **6.213 V**

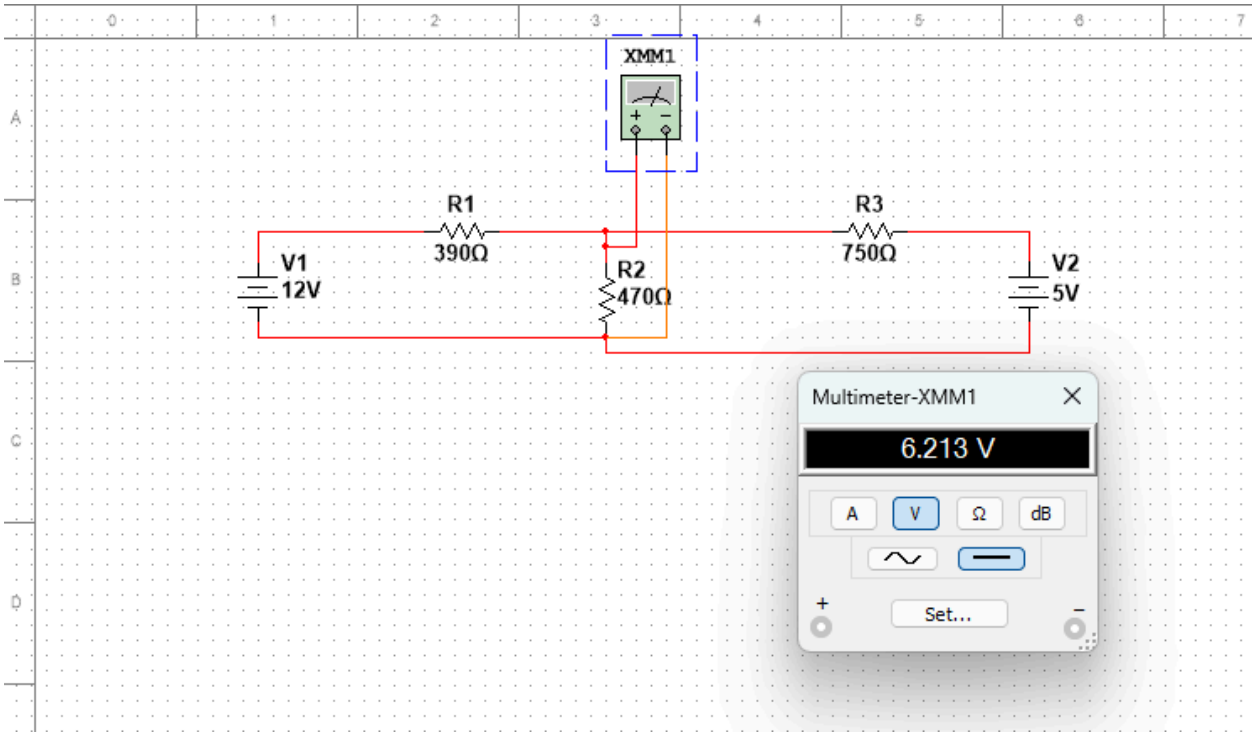
Check (both sources on):
V2 (1/390 + 1/470 + 1/750) = 12/390 + 5/750
V2 (0.006025) = 0.037436, so V2 = **6.213 V**

Part II. Circuit Simulation Using Multisim

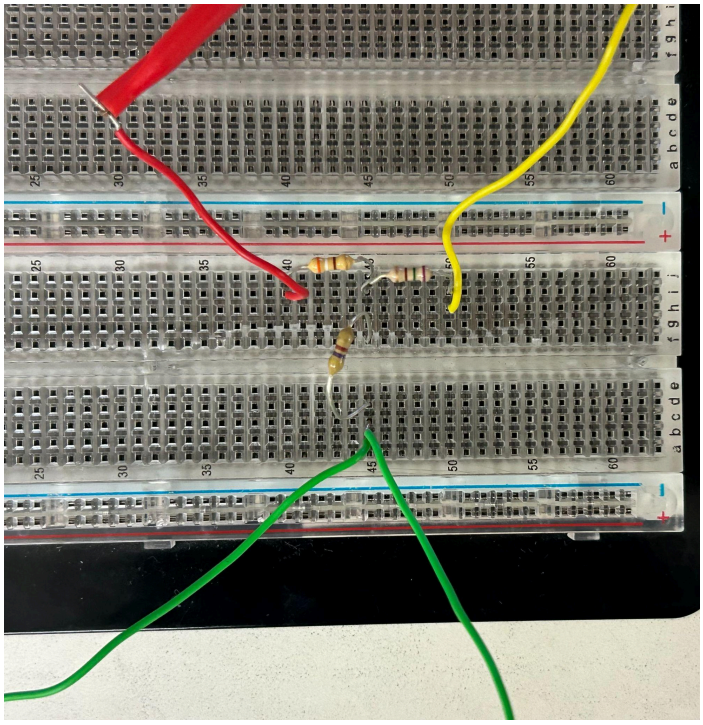


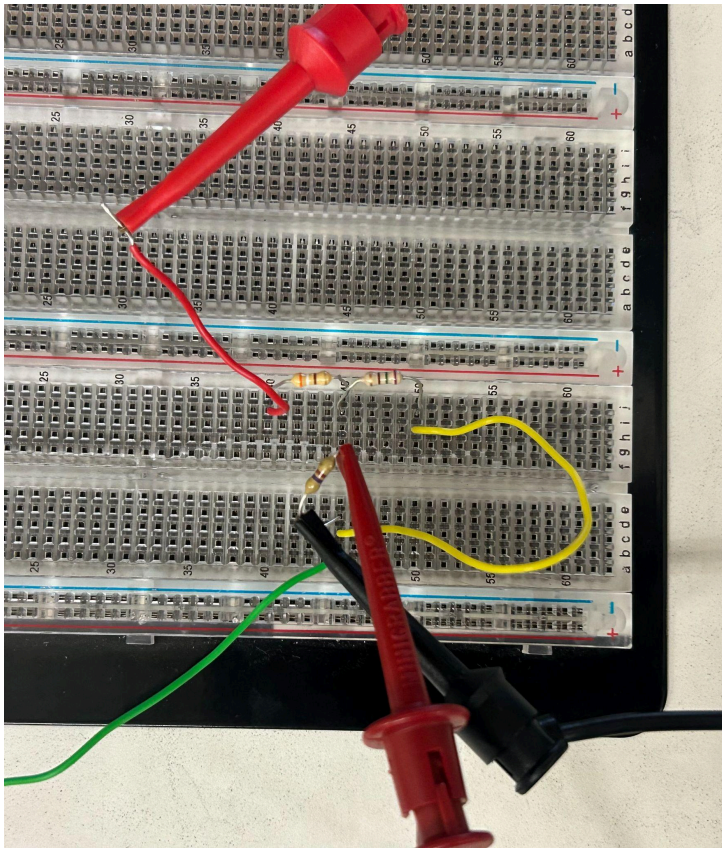


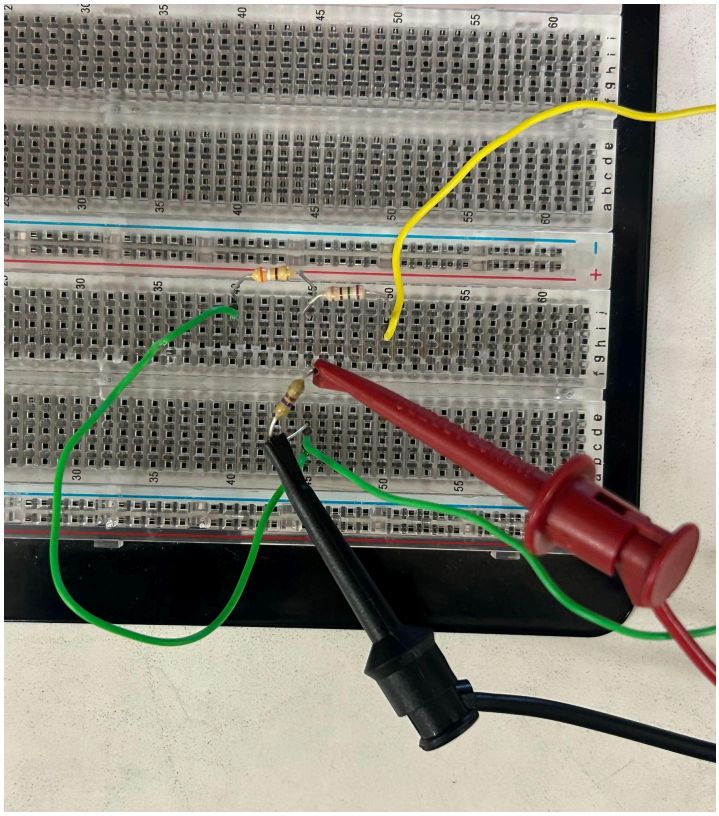




Part III. Circuit Construction on the Breadboard








KEYSIGHT
EDU34450A
5 ½ Digit Multimeter
Auto Trigger
DC Voltage
EDU34450A
KEYSIGHT TECHNOLOGIES
DCV
Range
Auto
Resolution
Slow
1.088 05
VDC
10MΩ
Auto
SLOW
1V
Input Z
10MΩ
Auto

Data Tables

Figure 1. Resistance Data

	R1	R2	R3
Expected value	390 Ω	470 Ω	750 Ω
Measured	384 Ω	462 Ω	751 Ω
%Error between expected value and measured	1.54 %	1.70 %	0.13%

Figure 2. Voltage Data

	V2,S1	V2,S2	V2	V2,S1 + V2,S2	%Error: difference between V2 and (V2,S1 + V2,S2)
Hand Calculations	5.107 V	1.106 V	6.213 V	6.213 V	0.00 %
Multisim	5.107 V	1.106 V	6.213 V	6.213 V	0.00 %
Breadboard	5.127 V	1.094 V	6.188 V	6.221 V	0.53%
%Error between hand calculations and breadboard	0.39%	1.08%	0.40%	0.13%	N/A

Percent Error Calculations

%Error = |Measured - Expected| / Expected × 100%

Questions

1. Compare the measured resistor values to the expected values. If they are not exactly the same then explain briefly why they are not exactly equal.
The measured resistor values are slightly lower than the expected values. THis is due to the internal resistance of the DMM, as well as the manufactured tolerance of the resistors that states that the value may range +/- slightly from the expected value.

2. What do you observe regarding the relationship between the voltages V2,S1, V2,S2, and V2 for the calculated case?
V2,S1 and V2, S2 do add up to V2. All 3 forms of verification (hand calcs, Multisim, and breadboard) confirm that the sum of the single-source voltages is equal to the combined

voltage. This proves the Superposition Theorem

3. What do you observe regarding the relationship between the voltages V2,S1, V2,S2, and V2 for the simulation case?
In Multisim, V2,S1 (5.107V) and V2,S2 (1.106 V) add up to the total V2 (6.213 V) exactly. THis leaves 0.00 % percent error

4. What do you observe regarding the relationship between the voltages V2,S1, V2,S2, and V2 for the breadboard case?
V2,S1 (5.127V) + V2,S2 (1.094V) = 6.221 V. This value is very close to the V2 that was measured at 6.188V. THe percent error here is only 0.53%. This confirms the Superposition Theorem as the V2,S1 + V2,S2 = V2.

5. Compare the measured and the Multisim values of voltages and explain any differences.
The multisim and hand calculations will have the same values since both are theoretical. The measured values from the breadboard were slightly different since the resistor values were not exactly the same, some being above 1% off. This resulted in our percent error between V2 and V2,S1+V2,S2 only varying by 0.53%

6. From your experimental results have you gained a good understanding of the Superposition Theorem?
Yes, we have gained a good understanding of the Superposition Theorem. We confirmed that V2,S1 + V2,S2 = V2 in three different ways. We used hand calculations to prove the theory, then applied it to Multisim to test the theory visually. Finally, we confirmed the theorem via a real breadboard circuit and found that the measured values are very similar (with some percent error due to internal resistance).

Challenges

One challenge in this lab was shorting the source on the breadboard. We had to remove the power supply and place a jumper wire in its place, and it took some time to ensure the short was connected to the correct nodes. Another challenge was caused by the new configuration of our circuit, which has two power sources. Because of this, our setup of the breadboard circuit took longer than normal.

Conclusion

In conclusion, we learned on how to do the sources on the breadboard, even when he had caused a few mistakes but we eventually fixed our mistakes. We learned by doing the hand calculations for the breadboard and designing the circuit stimulation. Learning how to do the resistors on the breadboard while one of them was doing a parallel circuit was a moderate task to do.